

Detection Quality Adapter

Object detectors fail in two directions, and both distort the measured yield of usable footage. Part one specifies a general adapter that corrects both. Part two supplies the parameters and additional rules that hands require.

PART ONE

1. The two failures

A **false positive** is a detection reported where the object is not present. A **false negative** is an object present in the frame that the detector does not report. False positives retain footage that should be discarded; false negatives discard footage that was sound. Neither is visible from a single frame, which is why both are corrected after detection rather than inside it.

2. What the adapter is

The adapter is a post-processing stage between a detector and everything downstream.

Contract	
Consumes	Per frame: a candidate pool of boxes with confidences, plus frame timestamps
Returns	Corrected detections, each marked as reported, merged, rejected or interpolated
Configured by	Per-class expectations: plausible size, plausible shape, candidate pool size and class maximum, maximum speed, and how the object typically leaves the frame
Does not	Alter the detector or require retraining. It does ask the detector to stop capping instances itself, and takes over that selection

The record of changes is not optional: an interpolated detection is an assertion the detector never made, and consumers must be able to tell the two apart.

3. Rejecting false positives

Six classes are separable without a model. The first three are **geometric** and need only the frame the box appears in. The last three are **temporal**: the box is individually plausible and is rejected only for how it behaves across frames. Order of application matters.

Order	Class	Basis	Basis for rejection
1	Duplicate detections	Geometric	Two heavily overlapping boxes on one object. Merged, retaining the stronger
2	Implausible size	Geometric	Far larger or smaller than the object can be at working distance
3	Implausible shape	Geometric	Markedly elongated where the object is not
4	Implausible displacement	Temporal	The box jumps further between consecutive frames than the object can travel
5	Unsupported detection	Temporal	Appears for one or two frames with no track before or after it
6	Static detection	Temporal	Does not move while the camera does, indicating scene structure

The split fixes where each runs. Temporal rules need tracks, which only exist after the association step in section 4, so the adapter runs geometric rejection, then association, then temporal rejection, then interpolation.

This is why the instance cap must be deferred rather than applied at the detector. A detector asked for the class maximum selects by confidence in a single frame, and that is exactly what is unreliable here: a duplicate or a background object can outscore a real one in a given frame, and the real detection is discarded before the adapter ever sees it.

The detector should instead emit a candidate pool slightly above the class maximum, and the adapter should select after association, ranking candidates by the track supporting them rather than by their score in one frame. A candidate on a continuous trajectory is kept; an isolated one is dropped as a flicker even where it scored well. This admits more false positives at the input, which is acceptable because the temporal rules exist to remove them, but the pool should stay narrow: a wide one only gives association more spurious candidates to confuse.

4. Recovering false negatives

Detectors intermittently drop an object that remains clearly visible. Frame by frame this is indistinguishable from absence; as a trajectory it is not, because a real object follows a continuous path and reappears close to where its motion predicted.

The adapter therefore associates detections into per-object tracks, extrapolates each across a break, and interpolates the missing frames when a track resumes near its predicted position. A break that does not resume is left untouched, and a detection far from any active track starts a new one.

The governing constraint is that an object leaving the frame is not a false negative. Filling that gap invents an object in empty footage and inflates the very yield the stage exists to measure. Before interpolating, the adapter tests whether the track was leaving: near a border with motion directed outward. If so, it ends there.

PART TWO

5. Specialising the adapter for hands

Hands are one configuration of the adapter above. Most of the specialisation is parameters rather than new logic, but four properties change behaviour materially.

Property	Consequence for the adapter
At most two instances per frame	The cap is low and constantly reached, so deferring it matters most here: a duplicate does not merely add an error, it displaces the subject's other hand. Request three or four candidates and select two after association
Head-mounted camera	The camera moves constantly, so the static-detection rule is unusually discriminating. A hand is never stationary in the image for a sustained period
Characteristic exit path	Hands most often leave downward, out of view below the camera, rather than sideways. The leaving-the-frame test must weight the lower border accordingly
A second eye is available	The wearer's hands are within arm's reach and a bystander's are not, so per-box stereo depth separates them directly. This is the strongest discriminator available and has no equivalent in the generic adapter, which sees one view

Two further properties are easy to assume incorrectly. Hands are roughly equant, so the shape rule is informative here where for elongated objects it would not be. And handedness is unreliable when hands cross, so tracking must associate on position and motion, not on the reported label.

6. Edge cases for hands

Case	Observation	Required handling
Duplicate boxes on one hand	Two heavily overlapping boxes, one hand present	Merge in the candidate pool, before selection down to two
Hand exits at a side border	Track ends near an edge, motion directed outward	Terminate the track. Do not interpolate
Hand passes below the camera	Track ends at the lower border, occluded by the torso	Terminate the track. Do not interpolate
Brief occlusion by an object	Track ends mid-frame, resumes near the predicted position	Interpolate the gap
Hands cross or overlap	Two genuine hands produce near-coincident boxes	Retain both. Must not be treated as a duplicate
Hand re-enters after a long absence	New detection distant from any active track	Start a new track rather than bridging
Motion blur during rapid movement	Confidence falls across several consecutive frames	Interpolate where the trajectory remains continuous
Another person's hands in view	Detections not belonging to the wearer, often entering from a border like a real hand	Reject beyond arm's reach by stereo depth. Position and entry direction alone do not separate them
Gloved or partially occluded hand	Reduced or intermittent confidence	Behaviour unmeasured. Needs labelled examples

Representative rather than exhaustive; more cases are expected once labelled data is reviewed.

7. Calibration

No value here should be set by inspection. A labelled reference set is required, weighted toward difficult cases rather than representative footage: both hands visible, one hand, no hands, hands at the frame border, and borderline detections. Frames containing no hands at all matter more than their share of footage suggests: a detector trained only on frames that contain hands has never been shown that the answer can be none, and will over-fire where it is.

Two quantities determine the configuration. First, the observed frequency of each false-positive class, so effort goes to the errors that actually occur rather than the ones easiest to describe. Second, the distribution of dropout lengths, which sets the largest break that may safely be interpolated: too low and genuine dropouts stay uncorrected, too high and the adapter interpolates across real absences and fabricates detections. The candidate pool width follows from how often a true hand is ranked below a spurious box, which the same reference set measures directly.

8. Validation

Report precision and recall per stage rather than in aggregate, because each stage should move a specific direction: false-positive rejection raises precision, trajectory recovery raises recall, and duplicate merging raises both since it frees an instance slot. Any stage moving the wrong way is misconfigured, and an aggregate figure conceals it.

Monitor the proportion of interpolated detections as a standing check. It is the clearest early signal that the adapter has stopped recovering objects and started inventing them, and needs no labelled data.